# A3-PS001-TC001

Android Ransomware-like Activity Early Warning in a Controlled Lab

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# Problem Description

- Android devices are at risk of ransomware attacks that can encrypt, modify, or make user files inaccessible.
- Detecting ransomware-like behavior early is challenging because suspicious file activities may resemble normal application operations.
- Existing detection approaches may struggle to identify unfamiliar threats, creating a need for behavior-based monitoring and early warning.
- This project proposes a controlled-lab prototype that analyzes file activity, classifies suspicious behavior into risk levels, and generates explainable alerts using safe synthetic or authorized test data.

# Background

- **Ransomware:** Malware that can encrypt or otherwise make files in accessible.
- **Detection Techniques:** Signature-based methods match known indicators; behavior-based methods analyze activity patterns.
- **Static and Dynamic Analysis:** Static analysis examines an application without execution; dynamic analysis observes behavior during execution.
- **Early Warning and Android Sandboxing :** Sandboxing limits access to other applications' private data, so the prototype must only claim visibility into events it can actually observe.

# Literature & RFC Review

| Research area | What existing research contributes |
|---|---|
| Early – stage ransomware detection | Investigates detecting ransomware before its activity progresses substantially. |
| Real-time behavioral detection | Studies behavior and system-call patterns for detecting ransomware. |
| Android malware ML and DL | Explores static, dynamic, and hybrid features for malware classification. |
| Explainable AI | Studies how to make malware-detection decisions understandable. |

# Proposed Solution

- **Test Scenario Generation:** Generate labeled normal and suspicious synthetic file-event sequences.
- **Event Validation:** Check timestamps, event types, required fields, and labels.
- **Feature Extraction:** Calculate operation counts, rates, and patterns in rolling time windows.
- **Detection engine :**Compare fixed-threshold rules with an improved detector; produce risk levels and explanations.
- **Dashboard and Evalution :** Display alerts, event timelines, detector comparisons, and downloadable results.

# Optimization Proposed

| Aspect | Basic Approach | Proposed Improvement |
|---|---|---|
| Detection | Rely only on known signatures or manual review | Examine observable behavior patterns |
| Explainability | Show only a detection label | Show event counts and reasons |
| Testing | Inconsistent manual examples | Repeatable synthetic test scenarios |
| False positives | Treat alerts as definitive | Test benign bulk operations too |
| Safety | Risky real- malware experiments | Harmless simulations in a controlled lab |

# Solution Architecture & Desian

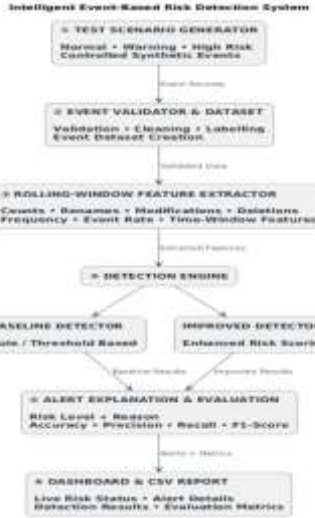



Figure: System Architecture UML Diagram

# Data Flow Diagram

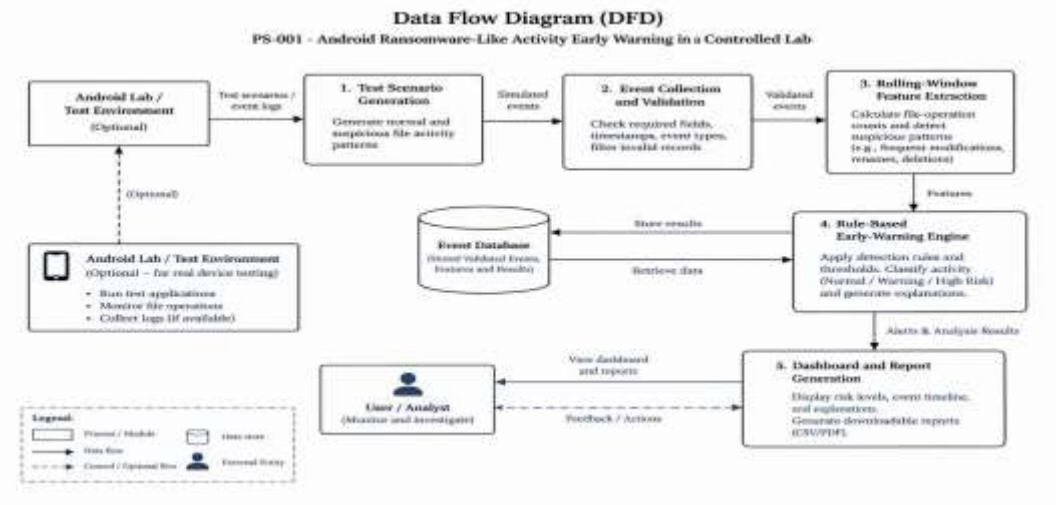



Figure: Data Flow Diagram (DFD)

# Deployment Timeline

| Stage | Work | Deliverable |
|---|---|---|
| 1 | Understand the problem and review related research papers and android security Documentation. | Requirements and references. |
| 2 | Design the system architecture, data flow and event schema. | Architecture diagram and DFD |
| 3 | Generate safe synthetic event data and implement the rule-based detection engine. | Working detection prototype |
| 4 | Develop a dashboard to display risk levels, alerts and detection results . | Interactive demonstration dashboard |
| 5 | Test normal and simulated suspicious activities and evaluate detection performance. | Evaluation results and limitations |
| 6 | Complete the project report, prepare screenshots, and record the team presentation video. | Final submission package |

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Thank You for Your Attention!